Dongho Lee

Center for Quantum Structures in Modules and Spaces,
Department of Mathematical Sciences, Seoul National University
Seoul, Republic of Korea
Homepage : https://dhlee-math.github.io
Mail : dhlee337@proton.me or tangled@snu.ac.kr

Research Interests

Keywords: Symplectic geometry; Floer theory; Hamiltonian dynamics; celestial mechanics.

My research focuses on Hamiltonian dynamics through the framework of symplectic geometry. More specifically, I study bifurcation phenomena of periodic Hamiltonian orbits using tools from Floer theory, with the goal of understanding how such local dynamical changes are reflected in symplectic and Floer-theoretic invariants. I am also interested in applications to celestial mechanics, especially restricted three-body problems and related Hamiltonian systems.

Employment

Postdoctoral Fellow May 2025 – Present
Center for Quantum Structures in Modules and Spaces (QSMS), Seoul National University

Postdoctoral Fellow Mar 2025 – Apr 2025
Research Institute of Mathematics, Seoul National University
Postdoctoral supervisor: Prof. Cheol-hyun Cho

Education

Ph.D. in Mathematics Mar 2018 – Feb 2025
Department of Mathematical Sciences, Seoul National University
Advisor: Prof. Otto van Koert

B.S. in Mathematics; B.A. in Economics; Minor in Physics Mar 2013 – Feb 2018
College of Liberal Studies, Seoul National University
Graduated with Honors

Publications and Preprints

1. **Mutations of Local Floer Chain Complexes across Hamiltonian Bifurcations**.
With Hong-Kwon Jo arXiv preprint, arXiv:2607.22178, 2026.

2. **Conley-Zehnder Indices of Spatial Rotating Kepler Problem**.
Journal of Topology and Analysis, published online, arXiv:2506.14325, 2026.

3. **Global Hypersurfaces of Section for Geodesic Flows on Convex Hypersurfaces**.
With Sunghae Cho *Archiv der Mathematik*, 123, 291–307, arXiv:2401.02667, 2024.

Works in Progress

1. **Fiberwise Convexity of the Circular Restricted Three-Body Problem**.
With Sunghae Cho and Beomjun Sohn. in preparation; expected Autumn 2026.

Thesis

1. **Global Hypersurfaces of Section and the Spatial Kepler Problem**.
Ph.D. thesis, Seoul National University, 2025.

Invited Talks

1. **Mutation of Local Floer Chain Complexes across Hamiltonian Bifurcations**.
   Universität Augsburg, Augsburg, Germany 1 Sep 2026
2. **Mutation of Local Floer Chain Complexes across Hamiltonian Bifurcations**.
   QSMS 2026 Summer Workshop, Jeju, Korea 10 Aug 2026
3. **Local Floer Chain Complex across Hamiltonian Bifurcation**.
   Geometry Space Surrey, University of Surrey, Guildford, UK 10 Jun 2026
4. **Conley-Zehnder Indices and Bifurcations of the Spatial Rotating Kepler Problem**.
   Dalian University of Technology, Dalian, China 19 May 2026
5. **Bifurcation of Periodic Orbits in Hamiltonian Systems**.
   G-Lamp Mathematics Seminar, Soongsil University, Seoul, Korea 22 Jan 2026
6. **Generic Bifurcation of Hamiltonian Orbits**.
   Symplectic Learning Seminar, Seoul National University, Seoul, Korea 12 Dec 2025
7. **Hamiltonian Dynamics and Three-Body Problem**.
   Rookie's Pitch, Seoul National University, Seoul, Korea 14 Oct 2025
8. **Three-Body Problem and Related Problems**.
   QSMS 2025 Summer Workshop, Goseong, Korea 21 Aug 2025
9. **Periodic Orbits of Rotating Kepler Problem**.
   Billiards and Quantitative Symplectic Geometry, Heidelberg, Germany 16 Jul 2025
10. **Closed Orbits of the Spatial Rotating Kepler Problem**.
    2025 Korean Mathematical Society Spring Meeting, Daejeon, Korea 25 Apr 2025

Poster Presentation

1. **Periodic Orbits and Symplectic Invariants in the Rotating Kepler Problem**.
   Celestial Mechanics & N-body Dynamics, Tokyo, Japan 16 Nov 2025

Selected Conferences and Workshops

Symplectic Topology - A conference in honor of Kai Cieliebak
Universität Augsburg, Augsburg, Germany 24–28 Aug 2026

Geometric Structures and Differential Equations – Symmetry, Singularity and Applications
RIMS, Kyoto University, Kyoto, Japan 13–17 Jul 2026

International Conference on Celestial Mechanics
SUSTech, Shenzhen, China 20–24 Apr 2026

New Developments in Symplectic Geometry
Seoul National University, Seoul, Korea 25–28 Nov 2025

East Asian Symplectic Conference 2025, Hokkaido University, Japan 25–29 Sep 2025

From Hamiltonian Dynamics to Symplectic Topology and Beyond
Institut Henri Poincaré, Paris, France 3–7 Jun 2024

Topology of 4-Manifolds and Related Topics Jeju, Korea 21–27 Jan 2024

East Asian Symplectic Conference 2023, Jeju, Korea 29 Oct–4 Nov 2023

Workshop on Symplectic Dynamics
Instituto Superior Técnico, Lisbon, Portugal 27 Jun–1 Jul 2022

Fukaya 60: Geometry and Everything, Kyoto University, Japan 17–22 Feb 2019

Teaching Experience

Teaching and Course Assistant, Seoul National University Mar 2018 – Feb 2025

- **Major and graduate courses:** Algebraic Topology (graduate), Introduction to Topology, Introduction to Differential Geometry, Selected Topics Seminar.
- **General mathematics courses:** Engineering Mathematics, Calculus, Calculus Practice, Calculus for Life Science, Fundamentals and Applications of Mathematics.
- **Head Teaching Assistant** for Introductory Mathematics Courses Spring 2020

Awards and Honors

Merit-based Scholarship
Department of Mathematical Sciences, Seoul National University 2018

National Scholarship, Korea Student Aid Foundation 2015

Merit-based Scholarship, College of Liberal Studies, Seoul National University 2013

Affiliations and Activities

Dynamical Systems: Limits and Recurrence, organizer 2026 – Present

Korean Mathematical Society, member 2025 – Present

QSMS-BK21 Symplectic Seminar, founding member 2021 – Present

References

Prof. Otto van Koert
Department of Mathematical Sciences
Seoul National University
okoert@snu.ac.kr
Ph.D. advisor

Prof. Cheol-hyun Cho
Department of Mathematics
POSTECH
chocheol@gmail.com
Postdoctoral supervisor